Baibhav Pratap Singh

Delhi, India | +91 7737731208 | b2p16s19@gmail.com
LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

Computer Science undergraduate with hands-on experience in **machine learning, AI systems, and data-driven application development** using Python. Built end-to-end ML pipelines for molecular property prediction and multi-agent AI systems, with experience in **Scikit-learn, Pandas, RDKit, FastAPI, and LLM-based applications**.

EDUCATIONS

Indian Institute of Information Technology Bhagalpur | Bhagalpur, Bihar — Aug 2023 – Jul 2027
Bachelor of Technology in Computer Science and Engineering

Akalank Public School (CBSE) | Kota, Rajasthan — Jul 2020 – May 2022
Intermediate

Delhi Public School (CBSE) | Hathras, Uttar Pradesh — Jul 2018 – May 2020
High School

WORK EXPERIENCE

Web Developer, The Realm of Six — Jan 2026 – Feb 2026

- Independently developed and deployed the official event website using Next.js and React, collaborating with stakeholders to deliver participant registrations, event timeline, and story-driven chapter navigation.
- Built **12+ reusable UI components** with responsive layouts and dynamic routing for seamless navigation across devices.

Technologies / Skills Used: *Next.js, React, JavaScript, Tailwind CSS*

PROJECTS

AI Drug Solubility Prediction | GitHub

- Developed an end-to-end machine learning pipeline for aqueous solubility prediction using the Delaney ESOL dataset, **cleaned to 1,123 drug-like molecules**.
- Engineered **2,056 molecular features** using RDKit descriptors and Morgan fingerprints, comparing **scaled Ridge, Gradient Boosting, and Random Forest regression** with five-fold cross-validation and **hyperparameter tuning of the selected Random Forest model**.
- Achieved **0.559 MAE, 0.822 RMSE, and 0.856 R^2** on a held-out test set and **implemented** an interactive Streamlit application for single-SMILES prediction.

Technologies / Tools Used: *Python, RDKit, Scikit-learn, Pandas, Streamlit*

Patheyatra AI – Multi-Agent AI Travel Planner | GitHub | Website

- Developed a multi-agent AI travel planner, coordinating **5 specialized agents** for intent analysis, destination research, weather forecasting, itinerary generation, and budget planning.
- Implemented asynchronous FastAPI workflows, enabling concurrent execution of independent agents and integrating multiple APIs for structured and unstructured data processing.
- Created interactive dashboards for itinerary visualization, budget insights, and trip management with **automated AI-generated travel plans**.

Technologies / Tools Used: *Python, FastAPI, Gemini API, Multi-Agent AI, React*

AI Appointment Scheduling Platform | GitHub | Website

- Developed an **AI-assisted appointment booking platform** using Next.js, TypeScript, PostgreSQL, and Supabase, automating booking workflows and scheduling decisions.
- Engineered scheduling algorithms for overlap detection, availability validation, slot generation, and conflict resolution to automate appointment decisions.
- Designed RESTful APIs and normalized PostgreSQL schemas, with **Supabase Realtime synchronization** to reduce booking conflicts and prevent double-booked concurrent users.

Technologies / Tools Used: *Next.js, TypeScript, Supabase, PostgreSQL, Tailwind CSS*

AWARDS & ACHIEVEMENTS

- Reached the finals of a national-level hackathon, **Hack4Bihar**, competing against teams from multiple institutions and **ranked 7th out of 1000+ participants**.
- Achieved a 3-star rating on CodeChef with a maximum rating of 1669.

POSITIONS OF RESPONSIBILITY

Event Lead and Host — Jan – Feb 2026

- Led an organizing team of **10+ members** to execute a campus-wide treasure hunt with registrations from approximately **400 participants** and a campus-wide offline finale featuring **30+ teams** (over **100 participants**) over two days.

Event Lead Team — Nov 2025

- Led cross-functional team coordination for event execution and on-ground operations, managing participant logistics and ensuring successful delivery for **800+ participants**.

SKILLS

ML / AI:	Scikit-learn, Pandas, RDKit, FastAPI, Streamlit, Gemini API
Programming Languages:	Python, C++, JavaScript, TypeScript, SQL, HTML5, CSS3
Databases:	PostgreSQL, MySQL
Tools / Engineering:	Git, GitHub, Linux, Docker, Postman, VS Code, React.js, Next.js, Node.js, Tailwind CSS
Soft Skills:	Problem Solving, Communication, Team Leadership, Collaboration, Adaptability

CERTIFICATIONS

- Code in Place – **Stanford University** | May 2025
- Career Essentials in Generative AI – **Microsoft and LinkedIn** | Jun 2025